Governing equations

unsteady Incompressible flow.

① conservation of mass.

$$\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} = 0.$$

② conservation of momentum equation:-

① x momentum equation:-

$$\frac{\partial u}{\partial t} + u\frac{\partial u}{\partial x} + v\frac{\partial u}{\partial y} = -\frac{1}{\rho}\frac{\partial P}{\partial x} + \frac{1}{\rho}\left[\frac{\partial}{\partial x}\left(\mu\frac{\partial u}{\partial x}\right) + \frac{\partial}{\partial y}\left(\mu\frac{\partial}{\partial y}\right)\right]$$

② y momentum equation:-

$$\frac{\partial v}{\partial t} + u\frac{\partial v}{\partial x} + v\frac{\partial v}{\partial y} = -\frac{1}{\rho}\frac{\partial P}{\partial y} + \frac{1}{\rho}\left[\frac{\partial}{\partial x}\mu\left(\frac{\partial v}{\partial x}\right) + \frac{\partial}{\partial y}\left(\mu\frac{\partial v}{\partial y}\right)\right]$$

③ μ → non-newtonian (shear thinning fluid flow).

The Bird - carreau method implemented,

$$\mu = \mu_\infty + (\mu_0 - \mu_\infty)\left[1 + \lambda(\dot{\gamma})^2\right]^{\frac{n-1}{2}}.$$

$\mu_\infty \rightarrow 0.0035$ Pa·s

$\mu_0 \rightarrow 0.056$ Pa·s

$\lambda \rightarrow 3.313$ s

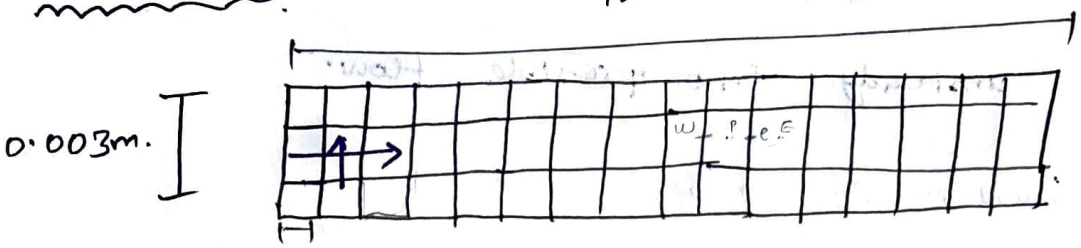
$n = 0.6$

$$\dot{\gamma} = \sqrt{\frac{1}{2}\Sigma_i \Sigma_j d_{ij} d_{ji}} \rightarrow \text{shear rate.}$$

$$\dot{\gamma} = \left[2\left[\left(\frac{\partial u}{\partial x}\right)^2 + \left(\frac{\partial v}{\partial y}\right)^2\right] + \left(\frac{\partial u}{\partial y} + \frac{\partial v}{\partial x}\right)^2\right]^{\frac{1}{2}}$$

Discretization:-

12cm = 0.12m.



$\Delta y = 1\text{mm} = 0.001\text{m}$

$\Delta x = 0.001\text{m}$.

$NX = 120$

$NY = 3$

$u_e > 0$.

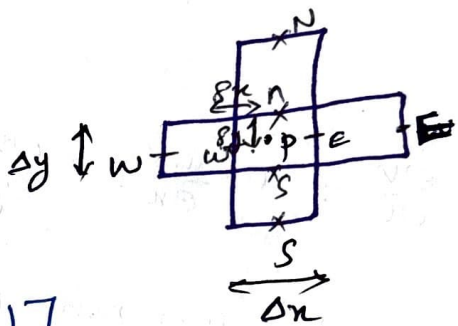
$v_n > 0$

Spatial discritization:-

Advection diffusion scheme:

general form.

$$\frac{\partial u u}{\partial x} = \frac{d}{dx}\, u_e \left(\frac{\partial u}{\partial x}\right).$$



① diffusion:- (x momentum)

$$\frac{\partial}{\partial x}\left(\mu \left(\frac{\partial u}{\partial x}\right) + \frac{\partial}{\partial y}\, \mu \left(\frac{\partial u}{\partial y}\right)\right]$$

$\phi = u$.

For 2D equation.

general form, $\frac{d}{dx}\left(\Gamma \frac{d\phi}{dx}\right)$.

$A_e = \Delta x$

$A_w = \Delta x$

$A_n =$

$$\left[\mu_e A_e \left(\frac{\partial u}{\partial x}\right)\right] = \frac{\mu_e A_e}{} \left[\frac{\phi_E - \phi_P}{(\delta x)}\right]$$

$$\left[\mu_w A_w \left(\frac{\partial u}{\partial x}\right)\right] = \mu_w A_w \frac{[\phi_P - \phi_w]}{\delta x}.$$



$$\left[\mu_n A_n \frac{\partial u}{\partial y}\right] = \mu_n A_n \frac{[\phi_N - \phi_P]}{\delta y}.$$

$$\left[\mu_s A_s \frac{\partial u}{\partial y}\right] = \mu_s A_s \frac{[\phi_P - \phi_s]}{\delta y}.$$

2D diffusion terms-

$$\left[\frac{\mu_e A_e}{\delta x}[\phi_E - \phi_P] - \frac{\mu_w A_w}{\delta x}[\phi_P - \phi_W]\right]$$

$$+\left[\frac{\mu_n A_n}{\delta y}[\phi_N - \phi_P] - \frac{\mu_s A_s}{\delta y}[\phi_P - \phi_S]\right]$$

general form derivatives. to x momentum,

$$\frac{\mu_e A_e}{\delta x}(u_E - u_P] = D_e[u_E - u_P]. \quad \left[D_e = \frac{A_e \mu_e}{\delta x}\right]$$

like for others also,

$$D_e[\phi_E - \phi_P] - D_w[\phi_P - \phi_W) + D_n(\phi_N - \phi_P) + D_s(\phi_P - \phi_S).$$

x momentum the Diffusion term,

$$\frac{1}{\delta}\left[D_e(u_E - u_P] - D_w(\phi_P - u_W) + D_n[\phi_N - u_P] + D_s[u_P - u_S]\right]$$

Y momentum equation.

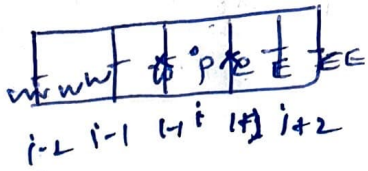
$$\frac{1}{\delta}\left[D_e[v_E - v_P] - D_w[v_P - v_W] + D_n[v_n - v_P] + D_s[v_P - v_S].\right]$$

Quick scheme For 1D momentum advection

Terms:

x momentum equation:-

$$\frac{\partial(uu)}{\partial x} + \frac{\partial(vu)}{\partial y}$$



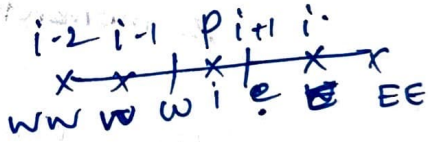
① East face:- $\frac{\partial uu}{\partial x}$ (direction of x)

General form of Quick scheme.

$$\phi_{face} = \frac{3}{8}\phi_{upwind} + \frac{6}{8}\phi_{central} - \frac{1}{8}\phi_{downwind}.$$

$$\phi_e = \frac{3}{8}\phi_E + \frac{6}{8}\phi_P - \frac{1}{8}\phi_W$$

$$\phi_w = \frac{3}{8}\phi_E + \frac{6}{8}\phi_P -$$



(or)

General form of Quick.

$$\phi_{face} = \frac{6}{8}\phi_{i+1} + \frac{3}{8}\phi_i - \frac{1}{8}\phi_{(i-1)}.$$

$$\phi_e = \frac{6}{8}\phi_P + \frac{3}{8}\phi_E - \frac{1}{8}\phi_W.$$

$$\phi_w = \frac{6}{8}\phi_W + \frac{3}{8}\phi_P - \frac{1}{8}\phi_{WW}.$$

General Form of Quick Scheme.

$$\phi_{face} = \frac{3}{8}\phi_{upwind} + \frac{6}{8}\phi_{central} - \frac{1}{8}\phi_{downwind}.$$

mathematically,

$$\phi_e = \frac{3}{8}\phi_E + \frac{6}{8}\phi_P - \frac{1}{8}\phi_W$$

$$\phi_w = \frac{3}{8}\phi_P + \frac{6}{8}\phi_W - \frac{1}{8}\phi_{WW}.$$

Substituting the u velocity.

$$u_e = \frac{3}{8}\phi_E + \frac{6}{8}\phi_P - \frac{1}{8}\phi_W.$$

$$u_w = \frac{3}{8}\phi_P + \frac{6}{8}u_W - \frac{1}{8}u_{WW}.$$

$$u_n = \frac{3}{8}u_N + \frac{6}{8}u_P - \frac{1}{8}u_S$$

$$u_s = \frac{3}{8}u_P + \frac{6}{8}u_S - \frac{1}{8}u_{SS}.$$

Substitute this equation in the x momentum equation.

$$F_e\left[\frac{3}{8}\phi_E + \frac{6}{8}\phi_P - \frac{1}{8}\phi_W\right] - F_w\left[\frac{3}{8}\phi_P + \frac{6}{8}u_W - \frac{1}{8}u_{WW}\right]$$

$$+ F_n\left[\frac{3}{8}\phi_N + \frac{6}{8}\phi_P - \frac{1}{8}u_S\right] + F_s\left[\frac{3}{8}u_P + \frac{6}{8}u_S - \frac{1}{8}u_{SS}\right].$$

$$= \left[\frac{3}{8}F_e\phi_E + \frac{6}{8}F_e\phi_P - \frac{1}{8}F_e\phi_W\right] - F_w\left[\frac{3}{8}\phi_P\right]$$

$$Fe\left[\tfrac{3}{8}\phi_E + \tfrac{6}{8}\phi_P - \tfrac{1}{8}\phi_W\right] - Fw\left[\tfrac{3}{8}\phi_P + \tfrac{6}{8}\phi_W - \tfrac{1}{8}\phi_{WW}\right]$$

$$+ Fn\left[\tfrac{3}{8}\phi_N + \tfrac{6}{8}\phi_P - \tfrac{1}{8}\phi_S\right] - Fs\left[\tfrac{3}{8}\phi_P + \tfrac{6}{8}\phi_S - \tfrac{1}{8}\phi_{SS}\right]$$

Solving the above equation.

$$\left[\tfrac{3}{8}Fe\phi_E + \tfrac{6Fe}{8}\phi_P - \tfrac{1}{8}\phi_W Fe\right] - \tfrac{3Fw}{8}\phi_P - \tfrac{6Fw}{8}\phi_W + \tfrac{Fw}{8}\phi_{WW}$$

$$+\tfrac{3Fn}{8}\phi_N + \tfrac{6Fn}{8}\phi_P - \tfrac{Fn}{8}\phi_S - \tfrac{3Fs}{8}\phi_P + \tfrac{6Fs\phi_S}{8} + \tfrac{1}{8}Fs\phi_{SS}\Big]$$

$$\tfrac{3}{8} = 0.375\,, \quad \tfrac{6}{8} = 0.75, \quad \tfrac{1}{8} = 0.125$$

Substituting the above equation.

$$[0.375]\,Fe\phi_E + [0.75]\,Fe\phi_P - [0.125\,Fe\phi_W]$$

$$- [0.375\,\phi_P Fw] - (0.75)\,Fw\phi_W + (0.125)\,Fw\phi_{WW}$$

$$+ (0.375]\,Fn\phi_N + (0.75\,Fn\phi_P] - (0.125)\,Fn\phi_S$$

$$- 0.375\,[Fs\phi_P] - 0.75[Fs\phi_S] + (0.125]\,Fs\phi_{SS}$$

$$[0.375\,Fe\phi_E] - [0.125\,Fe\phi_W] - 0.75(Fw\phi_W) + [0.125\,Fw\phi_{WW}]$$

$$+0.375\,Fn\phi_N - 0.125\,Fn\phi_S - 0.75\,Fs\phi_S + 0.125\,Fs\phi_{SS}$$

$$+ (0.75\,Fe\phi_P - 0.375\,\phi_P Fw) + [0.75\,Fn\phi_P]$$

$$[-0.375\,Fs\phi_P]$$

$$3/8\ Fe\ \phi_E - 1/8\ Fe\ \phi_W - 6/8\ Fw\ \phi_W + 1/8\ Fw\ \phi_{WW}$$

$$+ 3/8\ Fn\ \phi_N - 1/8\ Fn\ \phi_S - 6/8\ Fs\ \phi_S + 1/8\ Fs\ \phi_{SS}$$

$$+ \left[6/8\ Fe - 3/8\ Fw + 6/8\ Fn - 3/8\ Fs \right] \phi_P$$

- correcting these equations to ϕ formation,

$$3/8\ Fe\ \phi_E - \left[1/8\ Fe + 6/8\ Fw \right] \phi_W + Fw/8\ \phi_{WW}$$

$$+3/8\ Fn\ \phi_N - \left[1/8\ Fn - 6/8\ Fs \right] \phi_S + 1/8\ Fs\ \phi_{SS}.$$

$$+ \left[6/8\ Fe - 3/8\ Fw + 6/8\ Fn - 3/8\ Fs \right] \phi_P$$

Now Substitute the 2D Diffusion term,

$$3/8\ Fe\ \phi_E - \left[1/8\ Fe + 6/8\ Fw \right] \phi_W + Fw/8\ \phi_{WW} + 3/8\ Fn\ \phi_N$$

$$+\left[1/8\ Fn - 6/8\ Fs \right] \phi_S + 1/8\ Fs\ \phi_{SS} + \left[6/8\ Fe - 3/8\ Fw + 6/8\ Fn - 3/8\ Fs \right] \rightarrow \phi_P$$

~~$1/8 \left[3\ Fe\ \phi_E \ldots Fw \right] \phi_W + Fw\ \phi_{WW} + 3\ Fn\ \phi_N$~~

$$= De\ \phi_E - De\ \phi_P - Dw\ \phi_P + Dw\ \phi_W + Dn\ \phi_N$$

$$- Dn\ \phi_P - \phi_P Ds + Ds\ \phi_S$$

$$\tfrac{3}{8} Fe \phi_E - [\tfrac{1}{8} Fe + \tfrac{6}{8} Fw] \phi_W + \tfrac{1}{8}[Fw \phi_{WW}]$$

$$+ \tfrac{3}{8} Fn \phi_N - [\tfrac{1}{8} Fn + \tfrac{6}{8} Fs] \phi_S + \tfrac{1}{8} Fs \phi_{SS}$$

$$+ [\tfrac{6}{8} Fe - \tfrac{3}{8} Fw + \tfrac{6}{8} Fn - \tfrac{3}{8} Fs] \phi_P$$

$$= De \phi_E - De \phi_P - Dw \phi_P + Dw \phi_W + Dn \phi_N$$

$$- Dn \phi_P - Ds \phi_P + Ds \phi_S$$

Rewriting the above equation in the ϕ name.

$$[\tfrac{3}{8} Fe - De] \phi_E - [\tfrac{1}{8} Fe + \tfrac{6}{8} Fw + Dw] \phi_W$$

$$+ \tfrac{3}{8} [Fn - Dn] \phi_N - [\tfrac{1}{8} Fn + \tfrac{6}{8} Fs + Ds] \phi_S$$

$$+ \tfrac{1}{8} Fw \phi_{WW} + \frac{Fs \phi_{SS}}{8}$$

$$= - [\tfrac{6}{8} Fe - \tfrac{3}{8} Fw + \tfrac{6}{8} Fn - \tfrac{3}{8} Fs] \phi_P$$

$$- De \phi_P - Dw \phi_P - Dn \phi_P - Ds \phi_S.$$

$$= - \tfrac{6}{8} Fe + \tfrac{3}{8} Fw - \tfrac{6}{8} Fn + \tfrac{3}{8} Fs$$

$$= [-\tfrac{6}{8} Fe - \phi_E] + [\tfrac{3}{8} Fe - Dw) - \tfrac{6}{8} Fn - Dw + \tfrac{3}{8} Fs - Ds]$$

ϕ_P.

Final Equation of advection-diffusion problem can be, simplified It,

$$a_e \phi_E = \left[\tfrac{3}{8} F_e - D_e\right] \phi_E$$

$$a_w \phi_W = -\left[\tfrac{1}{8} F_e + \tfrac{6}{8} F_w + D_w\right] \phi_W$$

$$a_n \phi_N = \left[\tfrac{3}{8} F_n - D_n\right] \phi_N$$

$$a_s \phi_S = -\left[\tfrac{1}{8} F_n + \tfrac{6}{8} F_s + D_s\right] \phi_S$$

$$a_{ww} \phi_{ww} = \tfrac{1}{8} F_w \phi_{ww}.$$

$$a_{ss} \phi_{ss} = \tfrac{1}{8} F_s \phi_{ss}.$$

$$a_p \phi_p = -\tfrac{6}{8} F_e - \phi_E + \left[\tfrac{3}{8} F_w - D_w\right] - \tfrac{6}{8} F_n - D_n + \tfrac{3}{8} F_s - D_s$$

$$a_p \phi_p = a_w \phi_W + a_e \phi_E + a_n \phi_N + a_s \phi_S + a_w \phi_{ww} + a_s \phi_{ss}$$

$$a_p \phi_p = \sum a_{nb} \phi_{nb}$$

Finite volume Formulation. For pressure.

The general Equation of FVM.

$$\nabla(\rho \vec{V} \phi) = S_\phi + \nabla(\tau_\phi \nabla \phi)$$

$$S_\phi = -\nabla p$$

we know that

$$a_p \phi_p = \sum_{nb} a_{nb} \phi_{nb}$$

$$\iiint_{CV} \nabla p \, dV = \iint_{CS} p \cdot dA = \frac{P_e - P_w}{\delta x}$$

$$= -\left[\left[\frac{P_E - P_P}{\delta}\right] - \left[\frac{P_P - P_W}{\delta}\right]\right] \qquad A_e = A_w = \frac{\Delta x}{\delta} = \frac{\Delta y}{2}$$

$$= -[P_E - P_W] A_e$$

In y.

$$a_p \phi_p = \sum_{nb} a_{nb} \phi_{nb} + [P_W - P_E] A_e.$$

$$a_p \phi_p = \sum_{nb} a_{nb} \phi_{nb} + [P_S - P_N] A_n.$$

Rewrite this as in velocity form,

$$a_p u_p = \sum_{nb} a_{nb} \phi_{nb} + [P_W - P_E] A_e.$$

$$a_p v_p = \sum_{nb} a_{nb} \phi_{nb} + [P_S - P_N] A_n.$$

The ℓt must satisfy

$$F_e - F_w + F_n - F_s = 0$$

$$\boxed{u_e A_e - u_w A_w + v_n A_n - v_s A_s = 0.}$$

~~Ubby.~~

~~[illegible]~~

SEMI Implicit method For pressure Linked Equation.

$$\frac{u^{n+1} - u^n}{\Delta t} + [\text{convection}]^n = -\frac{1}{\rho}\left[\frac{\partial p}{\partial n}\right]^{n+1} + \nu\,[\text{diffusion}]^n$$

$$\frac{u^* - u^n}{\Delta t} + [\text{convection}]^n = -\frac{1}{\rho}\left[\frac{\partial p}{\partial x}\right]^n + \nu\,[\text{diffusion}]^n$$

From the Finite volume method.

$$a_e u_e = \sum_{nb} a_{nb} u_{nb} + (P_w - P_e) A_e \quad \rightarrow (1)$$

$$a_e v_e = \sum_{anb} a_{nb} v_{nb} + [P_s - P_N] A_n \quad \rightarrow (2)$$

$$F_e - F_w + F_n - F_s = 0$$

$$\rho u_e A_e + \rho u_w A_w + \rho v_n A_n + \rho v_s A_s = 0 \quad \rightarrow (3)$$

prediction step:-

$$a_e u_e^* = \sum_{nb} a_{nb} u_{nb}^* + (P_w^* - P_e^*) A_e \quad \rightarrow (4)$$

$$a_e v_e^* = \sum_{nb} a_{nb} v_{nb}^* + [P_s^* - P_N^*] A_e. \quad \rightarrow (5)$$

① − ④

$$a_e (u_o - u_e^*) = \sum_{nb} a_{nb} (u_{nb} - u_{nb}^*) + [P_W - P_E] A_e$$

→ ⑥

$$a_n (v_n - v_n^*) = \sum_{nb} a_{nb} (v_{nb} - v_{nb}^*) + [P_S - P_N] A_n$$

$$- [P_{S0}^* - P_N^*] A_n]$$

→ ⑦

PRIME Algorithm.

1. Compute the stresses.
2. Compute the velocities
3. Compute the pressure and correct velocity.
4. Compute the stress using the new velocities.
5. Compute the pressure again. Correct the velocity.
6. Return to step 1 again until convergence.

Discritization of Equations:-

① Cell size = 1mm = 0.001m.

speed u = 0.39 m/s

time step = 0.0001

$$CFL = \frac{u \Delta t}{\Delta x} = \frac{0.39 \times 0.0001}{0.001}$$

$$\boxed{CFL = 0.039}$$

② Discritization of the shear rate calculation.

$$\dot{\gamma} = \left[2 \left[\left(\frac{\partial u}{\partial x}\right)^2 + \left(\frac{\partial v}{\partial y}\right)^2 \right] + \left[\frac{\partial u}{\partial y} + \frac{\partial v}{\partial x} \right]^2 \right]^{1/2}$$

$$\frac{\partial u}{\partial x} = \frac{u_{i+1j} - u_{ij}}{\delta x}$$

$$\frac{\partial v}{\partial y} = \frac{v_{ij+1} - v_{ij}}{\delta y}$$

$$\frac{\partial u}{\partial y} = \frac{u_{i+1} - u_{i-1}}{2\delta y}$$

$$\frac{\partial v}{\partial x} = \frac{v_{ij+1} - v_{ij-1}}{2\delta x}$$

$$\left[\frac{\partial u}{\partial y} + \frac{\partial v}{\partial y} \right] = \left[\frac{u_{i+1j} - u_{i-1j}}{2\delta y} + \frac{v_{ij+1} - v_{ij}}{2\delta x} \right]^2$$

$$\dot{\gamma} = \left[2 \left[\left[\frac{u_{i+1j} - u_{ij}}{\delta x} \right]^2 + \left[\frac{v_{ij+1} - v_{ij}}{\delta y} \right] \right]^2 \right]$$

Boundary conditions:-

* Inlet Velocity (u) = 0.39 m/s.

* wall boundary condition = No slip
  Velocity = 0

* outlet => Pressure outlet set to be zero.
  Neuman boundary condition.

mass conservation:-

$$\int_{VP} (\nabla u) dV = \sum_f u_f S_f = 0.$$

For the 2D cartisian,

$$(u_e A_e - u_w A_w) + (v_n A_n - v_s A_s) = 0.$$